

Practical Work 5: Sequence Modeling and Attention Mechanisms

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Part 1: Mastering Basic Attention

1.1 Theoretical Background

Scaled Dot-Product Attention

The mathematical formula for Scaled Dot-Product Attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

Why is the scaling factor $\frac{1}{\sqrt{d_k}}$ necessary?

The dot products of queries and keys can grow large in magnitude, especially as the dimension d_k increases. Large dot products push the softmax function into regions where it has extremely small gradients (saturation). Scaling by $\frac{1}{\sqrt{d_k}}$ counteracts this effect, ensuring the gradients remain stable during backpropagation, thus facilitating effective training.

Self-Attention vs. Cross-Attention

The difference lies in the source of the input vectors:

- **Self-Attention:** The Queries (Q), Keys (K), and Values (V) all originate from the **same** source sequence. For example, in an encoder, each token attends to all other tokens in the same input sentence to capture internal dependencies.
- **Cross-Attention (Encoder-Decoder Attention):** The Queries (Q) come from the **decoder** (target sequence) layer, while the Keys (K) and Values (V) come from the **encoder** (source sequence) output. This allows the decoder to focus on relevant parts of the input sequence while generating the output.